Reasoning:

This lesson was designed to have students consider how evidence gathered via high quality methodology can be utilized to build scientific simulations. Students will identify high quality evidence that should be considered when making a health related decision and building scientific simulations. Students will apply their list of criteria for good methods to differentiate between high quality and low quality evidence. The process will show students that skills they utilized throughout the unit mimic those utilized by the scientists that created the simulation that they used.

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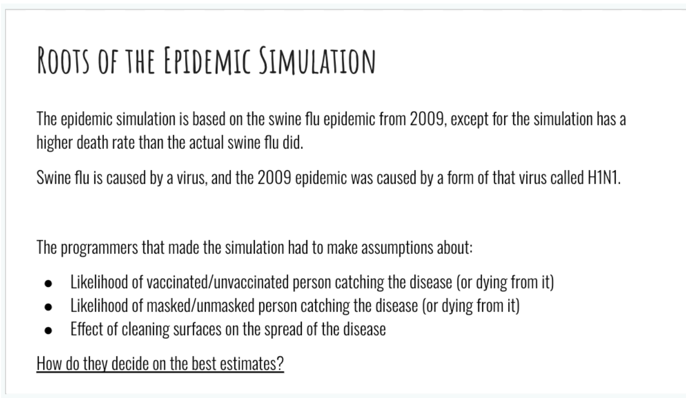

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🌎 The epidemic you read about is based on a real epidemic in 2009—the swine flu epidemic. Swine flu is caused by a virus. The 2009 epidemic was caused by a form of the swine flu virus that was called H1N1 to distinguish it from other swine flu virus.

How do they decide what the best estimates are?

Possible student responses:

- “They probably used data from the actual 2009 swine flu outbreak”
- “They could look at studies about how effective vaccines or masks are”
- “They might compare different studies and take an average or range”
- “They could test the simulation and adjust the numbers until it behaves like what actually happened”
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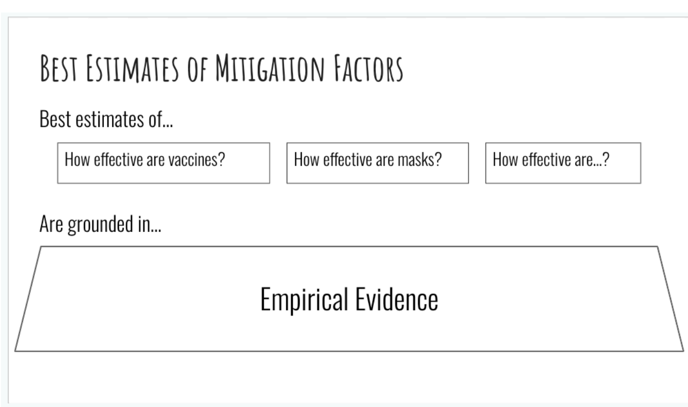

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The simulations are in fact grounded in empirical evidence.

🔬Scientists use empirical evidence to determine how effective vaccines, masking, and other mitigations are. The empirical evidence provides the best estimates of how effective the swine flu vaccine was, how effective masks were, etc.

Thus, we determine the relative efficacy of different mitigations from empirical evidence.
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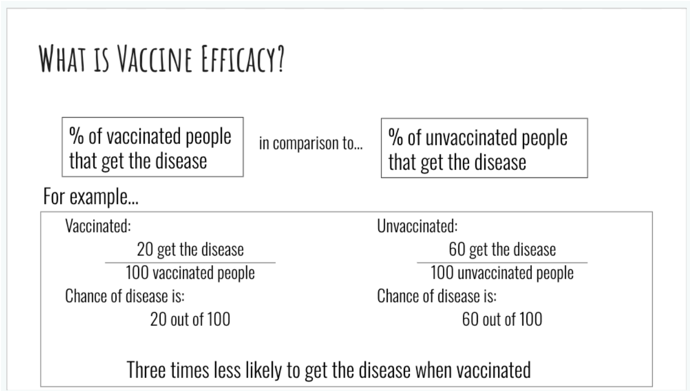


What is vaccine efficacy?

Vaccine efficacy tells you what percent of people get the disease after they get vaccinated, in comparison to the percent of the people who get the disease if they do not get vaccinated.

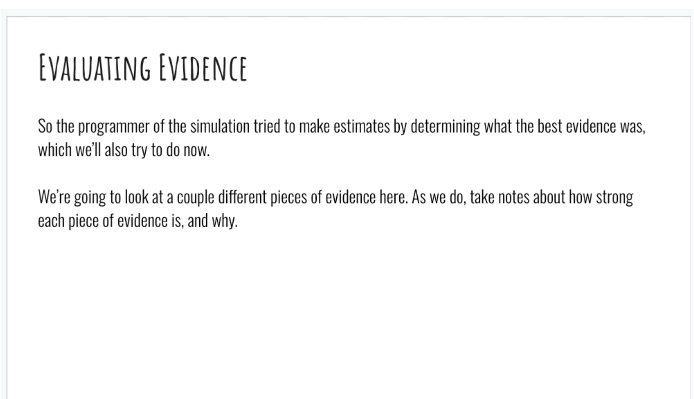
Here's an example:

Suppose there is a disease with a vaccine.

There are 100 people who do get vaccinated, and 20 of these people get the disease.

There are 100 people who do not get vaccinated, and 60 of these people get the disease.

This means that 20 vaccinated people got the disease instead of 60 people. This means that the chance of getting the disease was reduced by 3 times. In this (made-up) example, vaccinated people are 3 times less likely to get the disease. That tells you how effective the vaccine is.

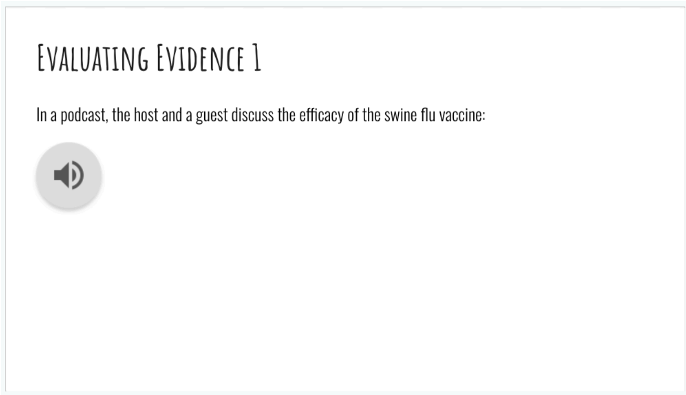
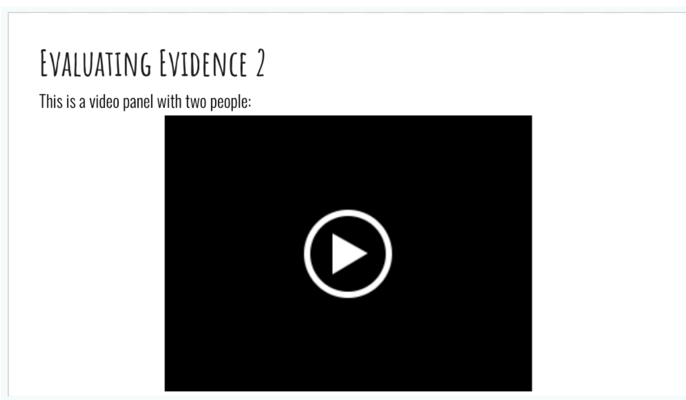


You can shorten this activity by removing either 1 or 2 pieces of evidence. If you remove 1 piece of evidence, remove the last one (5).

The programmer of the simulation used the best estimates of vaccine efficacy of the swine flu vaccination to determine how to program the simulation. What are the best estimates?

So let's look at some evidence for the efficacy of the vaccine for swine flu (H1N1).

🔍Evaluating the quality of evidence: Here are five pieces of evidence. As you go along, take notes about how strong each piece of evidence is, and why.

	Evidence 1: This is anecdotal evidence of a sort that is not reliable, because vaccines are not perfectly protective, nor do all unvaccinated people get the disease. So a few examples of individual people doesn't tell us anything. Another important feature of this evidence is that the guest is extremely inconsistent in the evaluation of evidence. The guest cites their own family as proof that the vaccine doesn't work, and then dismisses the host's counterexample by saying you can't tell from one family. Some students are likely to notice this, so call it out together as really bad reasoning in the class.
	Evidence 2: This is a single study—fine as far as it goes, but just one study. [Transcript]()
	Evidence 3: This is the best evidence because it synthesizes estimates from 34 studies. In addition, they gave greater weight to studies with better methodologies.

Evaluating Evidence 4 This is an excerpt from an article in a health magazine: But to understand the significance of this, it's important to first understand how effective the swine flu vaccine actually is. There are plenty of conclusions on this that you might find, but as always, we turn to findings from scientists. There was a recent study that took place in New York City, in which over 1000 people participated. The study estimated the efficacy of the vaccine by finding out whether or not people got the vaccine, and seeing if they caught swine flu within the study period. Of the vaccinated people, 12% got swine flu in this study. Of the unvaccinated people, 39% got swine flu. This means that getting the vaccination reduced the chances of getting the disease by 3.25 times.	Evidence 4: This is another perfectly fine study—not as strong as an experimental study, but one valuable kind of study for determining vaccine efficacy. But it is just one study.
Evaluating Evidence 5 This is a poll someone posted on social media: VaxFax @vaxfax · Sep 16 Hey, peeps 😁 I want to know what you 🤔 think about the efficacy of the swine flu vaccine. How much does taking the vaccine reduce your chances of getting swine flu? #swineflu #gettingthefax 24% Not effective at all. Just as likely to get flu when vaccinated. 35% Two times less likely to get the flu if vaccinated. 21% Three times less likely to get the flu if vaccinated. 12% Four times less likely to get the flu if vaccinated. 8% Five times less likely to get the flu if vaccinated. 1,956 votes · Final results	Evidence 5: This is not a study of actual vaccine outcomes, but simply an opinion poll which is not likely to be very correlated with actual outcomes.
Ranking Evidence & Forming Conclusions 1. Work with your group to rank each piece of evidence according to how strong the evidence is. You can use the "Ranking Evidence" tab in your experiment report. 2. Try to come to a consensus within your group 3. If you don't agree, discuss your reasoning 4. Then try to answer these questions: What is your best conclusion about how effective the H1N1 vaccine is? What are your reasons? How certain/uncertain are you?	Present all the evidence to the students, and then direct them to the last tab of their experiment report. The tab summarizes very, very briefly each piece of evidence and has a box for the ranking. Students rank the evidence from best to worst, working in small groups. 🔍 They should attempt to come to consensus, and try to persuade each other when they do not initially agree. Then hold a class discussion, and seek to bring out the strengths and weaknesses of the studies as indicated in the notes above. An extremely important point is that because the study in Evidence 3 reviewed all published research, it follows that the studies described in Evidence 2 and Evidence 4 are also included in Evidence 3. 🔍 The most important point is that it is better to rely on all the evidence to make scientific judgments, and not just on one or two studies. 🔍 A secondary but also important point is that it makes sense to give more weight to studies that are

	better methodologically.